

4(a)	1 less / no, ATP made ; 2 (so) poor muscle contraction / AW or (so) reduced impulse transmission ;	2
4(b)	any three from: 1 different children have different amounts of mutant, mtDNA / mitochondria ; 2 only mother / siblings, above threshold of mutant, mtDNA / mitochondria, will show symptoms ; 3 <i>ref. to</i> not showing typical patterns of inheritance ; 4 (mutant) mtDNA / mitochondria, inherited maternally or males do not pass on (mutant), mtDNA / mitochondria ;	3

Question	Answer	Marks
4(c)	<i>any two from:</i> <i>arguments for</i> 1 more diagnosis, means better management of those with the deficiency ; 2 can prevent losing further children to the deficiency / can decide to not have children / can choose to have children after embryo testing ; 3 <i>ref.</i> to possibility of using enucleated donor egg for mutated mtDNA ; <i>any two from:</i> <i>arguments against</i> 4 different opportunities for people to access the test ; 5 no cure ; 6 some people person may still lead a normal life ; 7 many mutations still unknown so may still not lead to positive diagnosis ;	3
4(d)	<i>any two from:</i> 1 change in, formation of α helix / secondary structure ; 2 change in, folding of / bonds in, tertiary structure ; 3 change in, 3D specific shape ;	2